

Drone and Payload Consideration

I. Introduction

The primary objective of this document is to describe the hardware requirements to fulfill the goals of the project. We look forward to your insights, criticism, and suggestions.

AirMetro – Project and Topic Introduction

- The AirMetro RTG (Research Training Group) takes a holistic approach to the vision of Air Taxis and Very low-level flying devices in Advanced Air Mobility (AAM). The RTG comprises 11 PhD projects, from T1 to T11, taking on challenges on different aspects like safety, trajectory optimization, navigation, economic feasibility, and mobility planning, as illustrated in Figure [1].
- Our goal for purchasing hardware equipment is to collect real-world data, utilize it to build our simulation models, and evaluate our algorithms on real data sets. The data collection and evaluation can potentially enhance collaboration in the group.

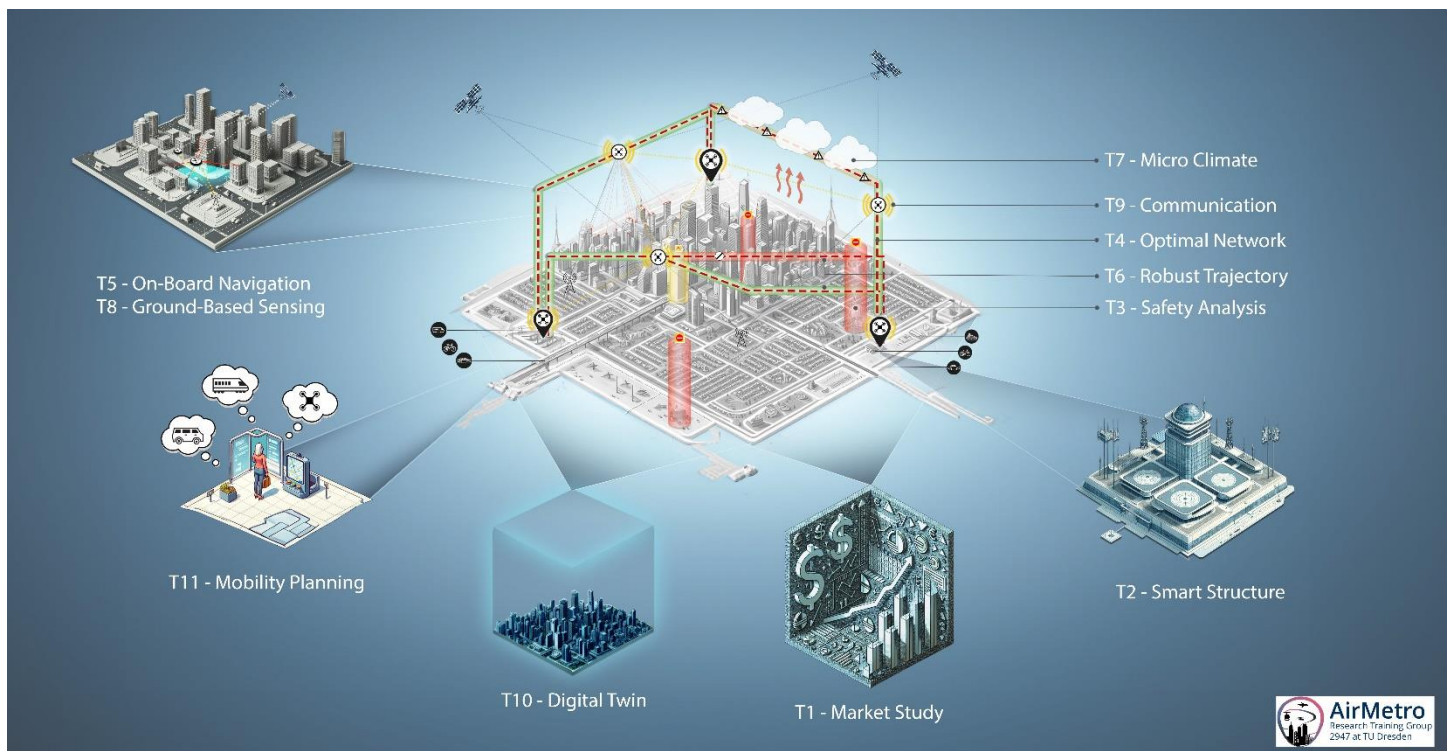


Figure [1]: RTG AirMetro Topic illustration

- **T1 – Market Study:** The aim of the topic is to develop and apply methods to economically analyze and evaluate decisions on AAM from a user, provider, and societal perspective and to identify policy

measures to regulate or promote AAM. A combined supply and demand decision on AAM will be modeled using approaches from microeconomic theory and discrete choice modeling, considering heterogeneous users, different technologies, infrastructure, usage and production costs.

- **T2 – Smart Structures:** The project aims to identify multi-physical phenomenon accompanying the incorporation of smart grid elements and other electronic devices within the AAM infrastructure that pose new challenges to robust functionality and safety of the entire system. State of the art numerical methods such as the Finite Element Method (FEM) and the Discontinuous Galerkin Time Domain (DGTD) method in the domains of Computational Electromagnetism and thermo-mechanics are employed and advanced to model and mitigate them.
- **T3 - Safety Analysis:** The aim of this topic is to create safe flight corridors up to the landing points and their safety-related design. The risk assessment of highly automated aircraft flying over populated areas is required for the approval and social acceptance of AAM. Different EU Regulations such as 2019/945, 2019/947 and 2019/317 follow a safety-based approach which examines the safety and its parameters for AAM operations. The risk assessment process includes understanding safety parameters in complex urban environments, safe flight procedures, and collision risk modeling.
- **T4 – Optimal Network:** This study aims to develop optimal network structures for AAM through the traffic management and spatial planning of landing sites (vertiports). In this process, multiple factors such as demand and technology integration, compatibility with existing infrastructure, safety regulations, economic efficiency, sustainability, and traffic flow management are considered.
- **T5 – On-board Autonomous Navigation:** The aim of this topic is to design a Simultaneous Localization and Mapping (SLAM) algorithm that runs on-board the aircraft and is integrated with the flight controller. The algorithm aims to deliver reliable position estimates from vision and inertial-based techniques, while also mapping the environment. The novelty lies in the semantic SLAM approach combined with graph optimization for pose estimates, landmarks, camera calibration, etc.
- **T6 - Robust Motion Trajectories:** This project aims to develop robust motion trajectories by quantifying polymorphic uncertainty. The uncertainty stems from environmental conditions, communication, and the inaccuracy of sensor and geodata. This facilitates the adaptive adjustment of the trajectory in real time based on the data and the models.
- **T7 – Microclimate:** The aim is to identify the weather-related operational boundaries for AAM. Take-offs and landings are influenced by the microclimate. Modelling of urban landscape to quantify the influence of adverse weather is essential. Thus, robust and precise weather forecasts in urban areas are a prerequisite. The weather phenomena are to be numerically simulated for specific UAVs in model cities.

- **T8 - Ground-based sensing:** For reliable positioning and navigation of a UAV, a new approach is to be developed that optimally fuses complementary navigation options (GNSS, INS, landmarks) in a situation-specific manner. Therefore, AI-based methods should be adapted to detect and match landmarks in acquired images based on various geodata. In addition to RGB images, the use of thermal images for detecting people and for use in low light conditions will be explored.
- **T9 - Communication:** The goal of this topic is to establish and evaluate reliable communication between UAVs and remote pilots at ground control stations, with a focus on terrestrial communication technology like 5G/LTE. It includes Command and Control communication and payload communication and explores the potential use of mesh networking for swarm operation. This topic also works alongside navigation risk assessments to build a Communication, Navigation, Surveillance (CNS) system for safe flight operations. UAVs will collect signal and traffic information to assess communication technology limitations and quality.
- **T10 – Digital Twin:** The goal of this topic is to ensure cyber-physical consistency in digital twin simulations for UAVs. Digital twins can analyze and predict complex traffic scenarios, but discrepancies between digital models and real-world conditions—such as weather variations—can disrupt system control and monitoring. This research focuses on methods to maintain continuous consistency through formal modeling, and context-aware simulations. By integrating sensor data and dynamically adjusting models in response to environmental changes, the study aims to accurately represent and adapt 4D flight paths, ensuring reliable AAM operations.
- **T11 – Mobility Planning:** Thanks to rapid technological development, travel via vertical take-off and landing (VTOL) aircraft such as air taxis could increasingly be considered a potential option for future passenger mobility. AAM offers several advantages, including low-congestion travel routes and higher speed compared to alternative modes on road and rail. However, to fully prepare for the emergence of AAM, methods for estimating future demand for this new mode of transportation need to be developed.

More information: <https://tu-dresden.de/bu/verkehr/airmetro/forschung/projects>

II. Experimental Setup

The Plan

The commercial UAVs are tailored to specific tasks with proprietary solutions making it non-ideal for research. Ambitiously designing UAVs from scratch is a mammoth and time-intensive task. Therefore, we seek a middle ground by customizing commercially available development kits with additional peripherals.

In the context of this document, our short-term goal is to obtain small and mid-sized UAVs (copter-based variants) to conduct initial testing.

UAV Equipment

We present a compiled set of core and topic-specific components. The shortlist provided is COTS UAV development kit, whose purpose is to integrate the components and conduct flight experiment for data measurements.

Core Components

Our goal is to utilize open-source tools, which align with the [Dronecode Foundation](#) standards. Therefore, we intend to use the [Pixhawk](#) flight controller with [PX4](#) autopilot software, [MAVLink](#) protocol, [ROS2](#) compatibility, and [QGroundControl](#) (QGC) as ground station.

Companion Computer

The higher-level tasks have to be offloaded from the flight controller to the companion computer. The following are our options: [Intel NUC](#), [Nvidia Jetson](#), or [Raspberry pi](#).

Camera

General requirements

- Cameras provide visual input for navigation in addition to the conventionally used IMU+GNSS
- High resolution images are required for semantic segmentation in **SLAM (T5)**, for the detection of **landmarks (T8)** and to accurately map the used air space (**T5, T8, T10**)
- Simplest setup is a Nadir-looking RGB-Camera
- Additional side-looking cameras (oblique) provide more comprehensive information (e.g. to map facades)
 - to capture multiple perspectives, a single oblique camera setup (1 nadir and 4 in the horizontal plane) was found to be another candidate, that would simplify the camera placement on the drone (Example: [Riebo D2M](#), [Riebo DG4M3](#))

- alternative: hemispherical cameras (fisheye cameras)
- Multiple cameras may be used in a stereo setting for distance information in addition to lidar
- A multi-view setup is relevant for safety in a future operating scenario
- Thermal cameras for flying in low light conditions, person detection, and additional information (features for matching algorithms)
 - cope with lower resolution with super-resolution methods

Connectivity and compatibility

- The considered connection to PX4 is via [MAVLink](#) or using a [camera manager](#) running on the companion computer
- The physical connection to the companion computer is possible via HDMI, ethernet, or through camera link using an external frame grabber ([PIXCI](#) or [iPORT CL-GigE](#)).
- A list of cameras (RGB and thermal) is available in [Cameras Compare.xlsx](#). An additional data sheet for oblique cameras is in [D2M DG4M3 Specification.xlsx](#).

LIDAR

- Depth information is vital for feature detection, obstacle avoidance and environment mapping which are important for SLAM (T5), identification of landmarks (T8) and for building a 3D map for digital twin (T10). The data can come from either stereo cameras or LiDAR.
- LiDARs as active sensors also provides illumination independent data
- It is proposed to use either:
 - 2D LiDAR: Light weight, cost-effective and builds the map along the flight, from the nadir perspective.
 - 3D LiDAR: Maps the complete terrain underneath the sensor including the depth along the heading direction.
- LiDAR data will be available for ROS via MAVLink.
- A small list of COTS LiDAR is in the appendix of this document.

Communication Terminal

- This payload is to establish reliable communication between the drone and the ground control station and provide internet connectivity (to access critical data like GNSS correction data).
- The data is divided into two main categories: [Command and Control \(C2\) Communication](#), which involves using the PX4 compatible MAVLink telemetry protocol to send critical mission information

like vehicle states (pose, speed, battery, etc.) and from GNSS, Barometer, IMU ensuring safe operation.

- Second, **Payload Communication**, which entails transmitting visual data (such as camera feed) LiDAR, and other essential data for safe drone operation to the ground station.
- The objective is to assess the communication link's performance when transmitting the information mentioned above between the drone and the ground control station.
- To ensure reliability, as the International Civil Aviation Organization (ICAO) suggests, multiple communication links are essential. Therefore, we plan to use Wi-Fi, 5G technology, along with LEO-based satellite communication in the future.
- The hardware equipment (terminal + antenna) is expected to weigh less than 700 - 800 grams.
- For 4G/5G communication terminal (modem) available in the below mentioned drone development kit. The use of SIM Card from service providers (such as Deutsche Telekom) to access the existing global (public) network.
- For better coverage area, the Wi-Fi links are planned to be equipped with Mesh software for leveraging swarm operation, from [Meshmerize](#) with [repeaters](#).
- If needed, private network based on [solutions](#) can be considered. But the purpose of private network must be re-evaluated to fit the vision of AirMetro.
- In the future, low-altitude flying-supportive LEO orbit satellite communication can be used. Although they offer less link capacity and higher latency, they are often considered a reliable backup communication option.
- The 600-1200 km LEO orbit latency fits the requirements of [ICAO RPAS](#) performance requirements such as link loss condition (exceeding 3 seconds). Therefore, we are considering using equipment based on Starlink and [Iridium](#).

Inertial Measurement Unit (IMU) sensor

- The IMU sensors are a crucial component of the minimum sensor setup needed by the chosen PX4 flight controller. 9-axis IMU sensors enable the tracking of the UAV motion, reliably, on a shorter time horizon.
- Considerations:
 - To minimize the IMU drift errors, some critical characteristics of the sensor come into play, namely the **instability and repeatability of bias and scale factor errors, and temperature sensitivity**.
 - Most COTS sensors qualify the requirements with their sufficient sampling rate, resolution, measurement range, sensitivity, and low cross-axis sensitivity or misalignment.

Hardware connection: Directly to the flight controller (I2C/SPI/Drone CAN)

RTK GNSS

- Positioning based on GNSS in combination with RTK is standard practice and allows for accuracy in the centimeter range.
- To ensure high accuracy, the receiver should support:
 - different GNSS (e.g. GPS, Galileo, BeiDou, GLONASS) -> multiple constellations
 - multi-band (faster convergence, larger baseline)
 - RTK algorithms that process correction data
- A suitable antenna (multi-band) is also of great importance
- Correction data can be received from reference stations (in Germany, we can use the DGNSS-Service SAPOS (<https://sapos.de/>)).
- PX4 supports the RCTM format (corrections) and SAPOS broadcasted RCTM via Internet (Ntrip) → it should be possible to transfer the required corrections to the GNSS receiver
- RTK GNSS receivers come with different specs. A product that uses a [ZED-F9P module](#) seems like a good choice.

The following equipment supports UAV operation.

1. Battery (LiPo)

- What are the specifications that need to be considered while selecting a LiPo Battery?
- From limited information based on online research, we consider [TATTU](#) and [Lumenier](#)
- Is purchasing rechargeable batteries and corresponding chargers a reasonable choice? (<https://www.fpvknowitall.com/fpv-shopping-list-charging/>). Your insights will be helpful as we lack comprehensive experience in battery and power distribution.

2. Sik telemetry radio

433 MHz is the standard used in Europe. Therefore, we consider [3DR](#) and [Holybro](#)

Potential Drone Development Kits

The following tables are a short list of 6 COTS drone development kits or kit families respectively.

Drone Features	HOLYBRO X650 Development Kit	HOLYBRO PX4 – X500 V2
URL	HOLYBRO-X650	HOLYBRO-X500-v2
Weight (actual with Batteries)	2800g	610g
Possible additional payload	Max. 3100g or 3500g – based on LiPo Battery weight	1500g (without battery)
Power (distribution)	XT60 plug for battery & XT30 plug for ESCs & peripherals	XT60 plug for battery & XT30 plug for ESCs & peripherals
Battery (flight time)	Approx. 30 min	Approx. 18 min (no additional payload)
Comm Link (Wi-Fi, 4G/5G, SiK Radio)	SiK Telemetry Radio	SiK Telemetry Radio
Additional information	Flexible rail mounting system available with and without GNSS	Flexible rail mounting system

Drone	QDrone 2	ModelAI Development Drones
URL	https://www.quanser.com/products/qdrone-2/#details	Various-ModelAI-UAVs
Weight (actual with batteries)	1500 g	265g – 1347g
Possible additional payload	Max 300 g	Up to 1000g for Sentinel
Power (distribution)	4S 14.8V LiPo (3700mAh) with XT60 connector	
Battery (flight time)	< 10 min	20 – 55 min
Comm Link (Wi-Fi, 4G/5G, SIK Radio)		Ghost atto (seems to require Win), 4G/5G, Wi-Fi
Additional information	Less information on website	VOXL2 companion computer Equipped with several sensors

Drone	Bitcraze Crazyflie 2.1+	UAV Systems Drones
URL	Crazyflie-2-1	UAV-Systems-Drones
Weight (actual with batteries)	27g	3712g - 12900g
Possible additional payload	15g	1500 - 11000g
Power (distribution)		
Battery (flight time)	7 min	25 – 70 min
Comm Link (Wi-Fi, 4G/5G, SIK Radio)	2.4GHz ISM band radio, Bluetooth	Telemetry Radio

Conclusion

In conclusion, this project is driven by specific goals (topic –wise) to provide potential solutions for AAM integration. We have carefully chosen the COTS hardware components for data collection that align with our objective, enabling us to commence the research promptly. Our current understanding has primarily focused on drone development kits. However, we are open to exploring and incorporating other types of drones that you may recommend. In addition, we have limited experience in drone payload deployment and operation. Therefore, your feedback and suggestions on our proposal will be invaluable in helping us refine and enhance our outcomes.

Appendix

A. LiDAR sensors under consideration:

- Garmin LiDAR-Lite v3
[GARMIN \(DE\) | LIDAR-Lite v3 | OEM | Sensoren - im Garmin Websop](#)
- LeddarTech Vu8
[Specsheet-Leddar-Vu-2.pdf \(leddartech.com\)](#)
- RPLIDAR A2M8 360°
[amazon.de/Slamtec-Scanning-Obstacle-Avoidance-Navigation/dp/B07T8XHJ9X](#)
- Velodyne HDL-32E (3D Lidar)
[97-0038_Rev M_HDL-32E_Datasheet_Web \(mapix.com\)](#)
- Velodyne Puck VLP-16 (3D Lidar)
[63-9229_Rev-H_Puck_Datasheet_Web \(goetting.de\)](#)
- OS2 OUTSTER (3D Lidar)
[datasheet-rev7-v2p5-os2.pdf \(ouster.io\)](#)
- OS1 (3D Lidar)
[Datasheets - Google Sheets \(ouster.io\)](#)
- Blickfeld – Cube 1 outdoor
[Cube 1 - Blickfeld with example \(LiDAR in Drones - Keeping an eye on things \(blickfeld.com\)\)](#)